World Happiness Report Visualization.

Final Assignment

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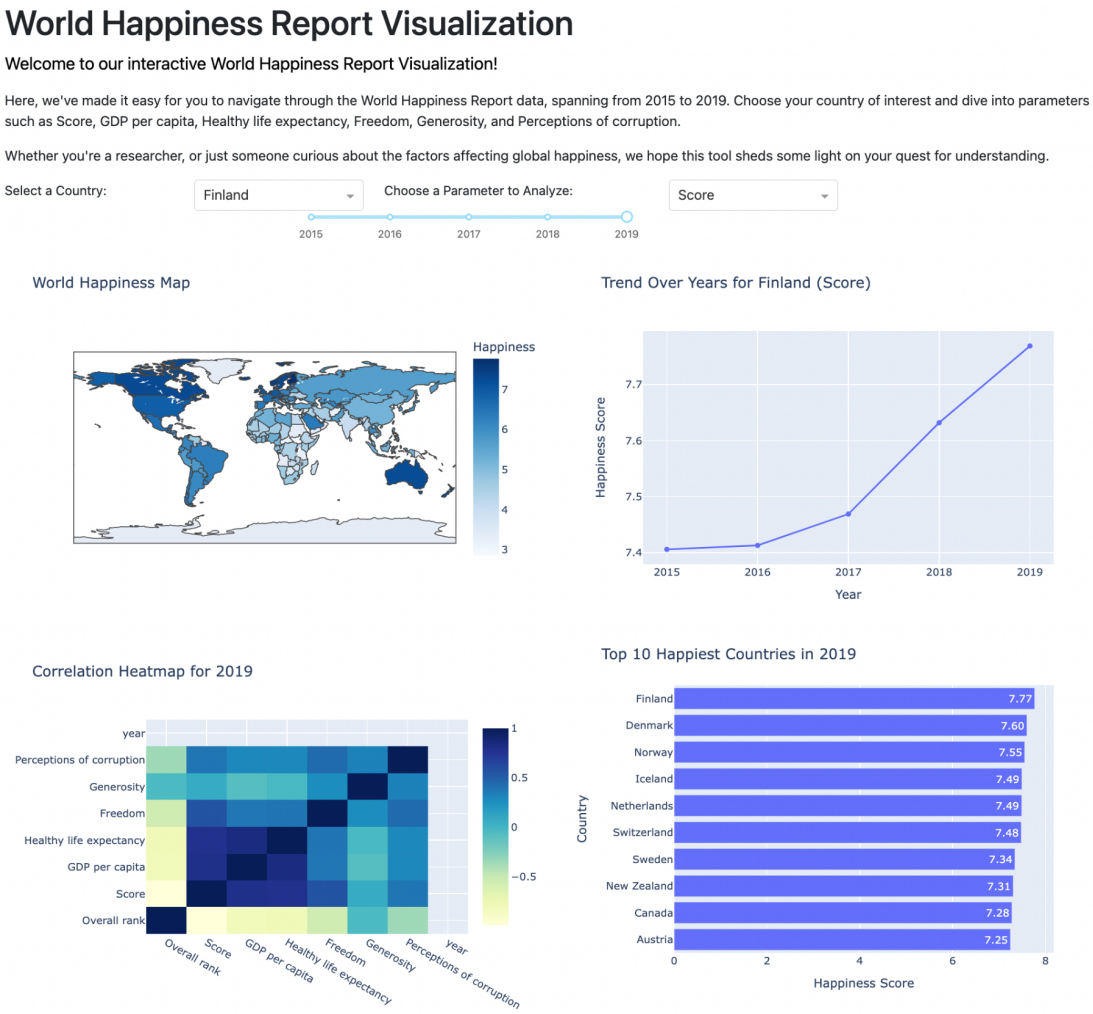
1. Overview



This visualization tool displays an interactive dashboard of the World Happiness Report data from the years 2015 to 2019. The dashboard includes a world map, a line graph showing the trend of selected parameters over the years for a specific country, a correlation heatmap, and a bar chart displaying the top 10 happiest countries based on their happiness score. Users can choose the country and specific parameters like score, GDP per capita, healthy life expectancy, freedom, generosity, and perceptions of corruption for detailed insight. Additionally, the dashboard also allows users to navigate through the data year-wise using a slider.

This script uses the Dash framework (a Python library for creating web applications) to create an interactive data visualization tool. The tool is designed to analyze the World Happiness Report data, which includes happiness scores for various countries from 2015 to 2019.

2. Data Preprocessing Pipeline

The script is, written in Python, and uses Dash, Plotly, and Pandas libraries to build an interactive web-based data visualization tool for the World Happiness Report.

This application processes World Happiness Report data from 2015 through 2019, each year's data stored in separate CSV files. It standardizes the datasets across these years, selecting important parameters such as happiness score, GDP per capita, healthy life expectancy, freedom, generosity, and perceptions of corruption.

Once the data processing is complete, it integrates these datasets into a comprehensive pandas data frame. The application's user interface allows users to explore this unified dataset in various ways.

This dataset is obtained from Kaggle.com. The link is as follows: https://www.kaggle.com/datasets/unsdsn/world-happiness

3. Features of Visualization

The main features of this Dash application include the following:

Interactive Dropdowns and Slider:

Country Selector: The application has an interactive dropdown that allows the user to select a specific country for analysis from the available options in the World Happiness Report data. This selection is used to focus the subsequent visualizations on the chosen country.

Parameter Selector: Another dropdown present in the application is the Parameter Selector. This allows the user to choose among various parameters like Score, GDP per capita, Healthy life expectancy, Freedom, Generosity, and Perceptions of corruption. This selected parameter is used to update the various charts and the world map visualization.

Year Slider: A slider that allows the user to select a year for the visualizations. This slider ranges from the minimum to the maximum year present in the World Happiness Report data, which, in this case, spans from 2015 to 2019.

World Happiness Map: This feature is an interactive world map that shows a choropleth representation of the selected parameter for the selected year. Each country's color intensity is proportional to the chosen parameter's value, providing a visual comparison of the parameter across different countries. Clicking on a specific country in this map will update the Trend Over Years visualization for the selected country.

Trend Over Years: This feature is a line chart that shows the trend of the selected parameter for a chosen country over the years. The trend line aids in understanding how the selected parameter has evolved over time for the specified country. The country for this visualization can either be selected from the dropdown or by clicking on the world map.

Correlation Heatmap: This is a heatmap that represents the correlation matrix between different parameters for the selected year. This visualization is crucial in understanding the relationships between different factors contributing to the happiness score. A high positive correlation between two parameters indicates that they tend to increase or decrease together, while a high negative correlation suggests that one parameter tends to increase when the other decreases.

Top 10 Happiest Countries: This feature is a horizontal bar chart showing the top 10 happiest countries based on the happiness score for the selected year. It provides a snapshot of the highest-ranking countries in terms of happiness. Clicking on a specific bar in this chart will update the country selector dropdown with the corresponding country, thereby allowing further exploration of the selected country's data in the other visualizations.

4. Design Rationale

The design of this Dash application aimed to facilitate an intuitive exploration of the World Happiness Report data. It caters to both technical and non-technical audiences, ensuring broad usability and accessibility.

The specific visual encodings, interactions, and animation techniques were selected to serve this purpose effectively.

Visual Encodings:

Choropleth Map (World Happiness Map): This visualization utilizes the natural geographical layout of countries to map the spatial distribution of data. Different color intensities correspond to different values of the chosen parameter (for example, happiness score), thus giving users a global perspective of the data. This way, the viewers can immediately grasp which countries perform better or worse for the selected parameter. It is especially effective in revealing patterns and trends related to geographical proximity or regional blocs.

Line Chart (Trend Over Years): The line chart is used to show the progress or regression of a particular parameter over a specific period. The X-axis represents time (years), and the Y-axis represents the chosen parameter. A line connects the points corresponding to each year's value, and its slope provides an intuitive grasp of the trend (rising, falling, or constant). The lines' color and thickness can be adjusted for multiple countries or parameters, creating an easy-to-interpret comparative view.

Heatmap (Correlation Heatmap): A heatmap utilizes color shades to represent numerical or categorical values. In the correlation heatmap, each cell's color intensity represents the degree and direction of correlation between two parameters. A dark color may indicate a strong correlation, while a light color may represent a weak correlation. It's an efficient method to visualize and detect correlations among multiple variables simultaneously, which can lead to insights into potential causes and effects or prompt further analysis.

Horizontal Bar Chart (Top 10 Happiest Countries): Bar charts are an easy-to-understand graphical representation that compares quantities among different categories, in this case, the happiness score among the top 10 happiest countries. The length of each bar corresponds to the happiness score, enabling users to make quick comparisons. The orientation (horizontal) makes it easier to read the country names and allows for better utilization of space.

Interactions:

Dropdowns and Slider: Dropdown menus for country and parameter selection provide a compact and organized way to present a large list of options. Users can intuitively navigate and select their desired options, thereby controlling the data displayed in the visualizations. The slider for year selection presents a range of options as a sliding scale, allowing users to swiftly navigate through time and observe the changes in the data.

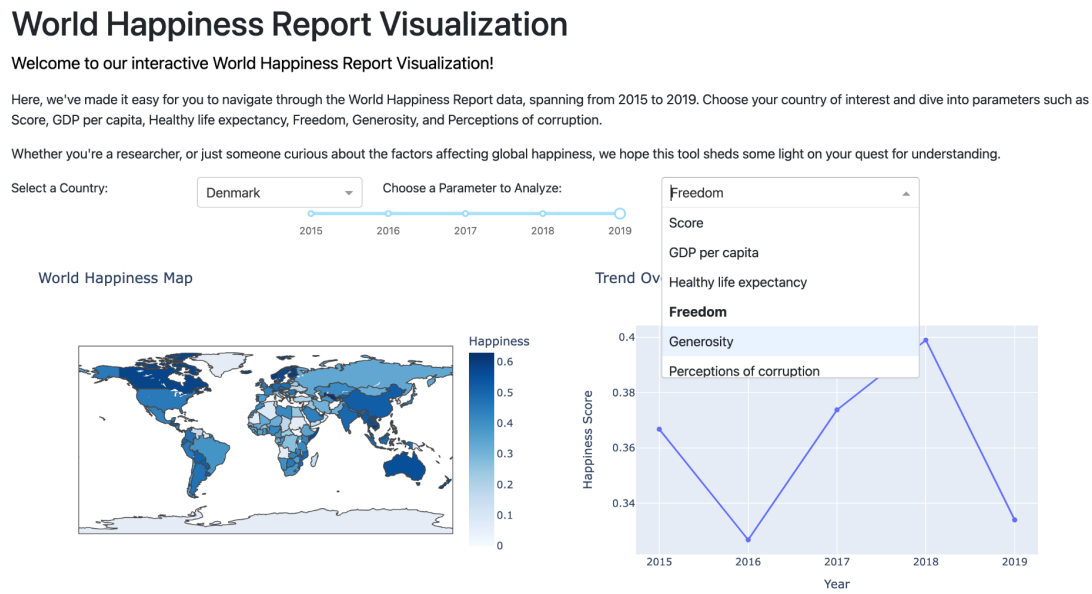
Graph Selection: Interactive graph selection allows users to delve deeper into the data by clicking directly on the visualizations. For example, by clicking on a country in the choropleth map or the bar chart, the line chart automatically updates to show that country's trend over the years. This direct manipulation interaction enhances the user's sense of control and engagement, providing a more immersive data exploration experience.

5. Usage Scenarios

The provided code is a Dash application that creates an interactive visualization of World Happiness Report data from 2015 to 2019. The application allows users to explore the data and identify trends and patterns across different countries and variables. The primary components of the application include a world map (Choropleth map), a line graph for trend analysis, a heatmap for correlation, and a bar chart for the top happiest countries.

How to use and explore the visualization:

Selecting a Country and Parameter: At the top of the visualization, users can choose a country of interest from the dropdown list and select a parameter (Score, GDP per capita, Healthy life expectancy, Freedom, Generosity, Perceptions of corruption) to analyze. The selected country and parameter are then reflected in the subsequent charts.



Navigating through Years: A slider allows the user to choose a specific year from 2015 to 2019. This affects all the plots, updating the data to reflect the year selected.

Viewing the World Map: This Choropleth map provides a global view of the selected parameter for the chosen year. The color intensity represents the magnitude of the parameter. Users can click on a country on the map to select it.

Analyzing the Trend Chart: The line graph shows the trend of the selected parameter over the years for the selected country. Users can click on a specific point in the line graph to select a year.

Interpreting the Correlation Heatmap: This heatmap visualizes the correlation among different parameters for the selected year. The color intensity represents the strength and direction (positive/negative) of the correlation.

Exploring the Bar Chart: This chart displays the top 10 happiest countries for the selected year based on the happiness score. Clicking on a country's bar in the chart will select that country.

Interactive Updates: The application is designed to allow interaction among the different plots. Selecting a country from the map or the bar chart updates the line graph, and vice versa. Also, clicking on a year in the trend line chart updates the year slider and all the other graphs for that year.

6. Observations Obtained from the System

The application provides four main visualizations:

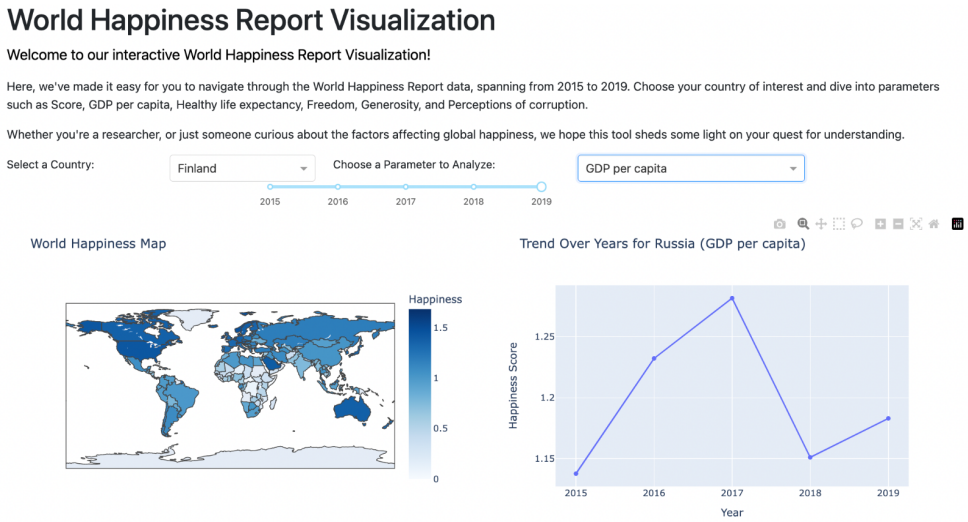
1. *World Map:* based on selected year and parameter
2. *Line graph* showing the trend over the years for a selected country and parameter
3. *Heatmap* showing a correlation between different parameters for a selected year
4. *Bar Chart* showing the top 10 happiest countries for a selected year

Users can interact with the visualization through a country selector dropdown, a parameter selector dropdown, and a year slider. Furthermore, clicking on a country in the world map or bar chart updates the line graph for that country. Clicking on a year in the line graph updates the world map, heatmap, and bar chart for that year.

Observations & Examples:

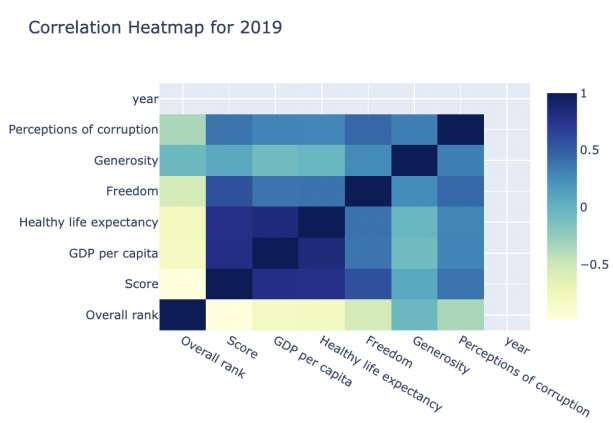
Trends Over Time: By selecting a country and a parameter in the dropdowns, you can observe how the chosen parameter changes over the years for the selected country. For example, if you

choose Finland and GDP per capita, you can observe how Finland's GDP per capita has changed from 2015 to 2019.



Geographical Differences: The world map gives a geographical representation of the selected parameter. For example, if you choose Happiness Score and slide the year to 2019, you can see a world map colored according to the happiness scores of countries in that year. This can reveal geographical patterns, such as certain regions being consistently happier than others.

Correlation between Parameters: The heatmap visualization shows the correlation between different parameters for a given year. This can provide insights into the relationships between these parameters. For instance, you may find that there's a strong positive correlation between GDP per capita and Happiness Score, suggesting that countries with higher GDP per capita tend to have higher happiness scores.



Top Performers: The bar chart visualization shows the top 10 countries with the highest happiness scores for a selected year. This can allow you to quickly identify which countries were the happiest in a given year.

7. Appendix: How to Run the Code

Running the code

To run the provided code, the necessary Python packages need to be installed in your Python environment. You can use the Conda environment, which is a package, dependency, and environment management for any language. You can create a new conda environment and install the required packages using the following steps:

First, create a new conda environment, let's call it "happiness_report", using Python 3.7:

conda create -n happiness_report python=3.7

Activate the environment:

conda activate happiness_report

Now, install the necessary packages inside this environment. You can use either pip or conda to install these packages. Here's an example of installing packages via pip:

pip install pandas dash plotly dash_bootstrap_components dash_core_components dash_html_components seaborn numpy plotly_express

Once all packages are installed, you can run the script using Python:

python infovis.py

Then follow the link on the terminal this will be generated: Dash is running on http://127.0.0.1:8050/